

A Possible Hybrid Machine Learning, BiLSTM, and Mass-Constrained Probabilistic Framework for Known-Size and Known-Mass Protein Scaffold Gap Filling

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Abstract—The protein scaffold gap-filling problem is a computational problem in de novo protein sequencing, where gaps contain unknown amino acids. We introduce a hybrid prototype for sampling scaffold gaps, which can predict missing scaffold gaps under either known-gap-size or known-gap-mass constraints. This study turned a synthetic protein sequence dataset into scaffold samples by stripping off subsequences and capturing their sizes, masses, and flanking contexts. A Random Forest baseline for first-residue prediction, a BiLSTM model to predict full-gap sequences, and a probabilistic mass-constrained k-mer retrieval model were implemented in three supporting branches. Then, the mass consistency, candidate frequency, and deep learning confidence were fused using a hybrid reranker. We test the framework with validation accuracy, token-level accuracy, exact-match accuracy, residue accuracy, and top-k recovery. Results demonstrate that local context alone does not fully recover candidates at small gaps on the synthetic dataset, and mass-constrained retrieval significantly helps recovery. The study demonstrates the feasibility of hybrid scaffold gap-filling and identifies homologous sequence retrieval and protein-language model embeddings as key directions for future improvement.

Index Terms—protein scaffold gap filling, de novo protein sequencing, mass spectrometry, machine learning, BiLSTM, probabilistic candidate retrieval, bioinformatics.

I. INTRODUCTION

Proteins are involved in most biological processes in living organisms, and understanding their function requires knowledge of their amino acid sequences [1]. Protein sequencing is an essential first step in understanding protein function, and the most widely used high-throughput method today is tandem mass spectrometry, which uses both top-down and bottom-up approaches to infer protein sequences from spectra [2]. However, these methods often yield incomplete sequences, or protein scaffolds - sequences with one or more contiguous gaps (or sequences) of amino acids that could not be determined directly from the spectral data. These gaps occur due to spectral noise, peak overlap, poor digestion, and the lack of highly similar sequences in sequence databases.

Computational gap filling is a key, unmet step in the de novo protein sequencing process [3], and has direct applications in antibody profiling, proteoform analysis, drug design, and vaccine design [4].

The problem is to predict as accurately as possible the amino acid sequence of one or more gap regions in a protein scaffold. There are two variants of this problem. In the known-gap-size variant, the gap size, for example, the number of amino acids to fill it, is known, enabling sequence-level prediction using local context features. In the known-gap-mass variant, the total mass of the missing piece is known. However, not its precise size, a situation that often arises in mass spectrometry when the mass of a peptide can be measured but the number of amino acids cannot. The goal is to predict an amino acid sequence, a score indicating confidence, and confirmation that the predicted sequence meets the size or mass constraints, for each gap. In the multi-gap case, a prediction must be made for all gap positions in a single scaffold.

The current methods approach gap filling in scaffold sequences using only one modeling approach, which has drawbacks [5], [6], [7]. Conventional machine learning methods, such as random forests and kNN, are fast but do not handle large gaps or rare motifs. Probabilistic methods address the known-mass problem by generating and scoring candidate sequences, but they require homologs to function. Deep learning models, for example, transformers and variants of GPT, capture sophisticated sequence patterns but may not obey mass constraints and require large datasets. There is no known system that combines these three approaches to create a hybrid that can handle both problem types and adaptively weigh the results. This motivates the proposed system, which hypothesizes that an integrated method that fuses machine learning predictions, probabilistic mass-constrained scoring, and deep sequence models will outperform any single method - especially in the face of noisy masses, lack of homologs, large gaps, and multiple-gap scaffolds. This paper has several

contributions as below:

- We propose the a novel hybrid approach for protein scaffold gap filling that is aware of both variants of the problem by a multi-branch hybrid prediction with three branches (traditional ML, probability scoring, and deep learning) and a learned meta-model to combine them.
- We build a feature pipeline that mixes encoded context features, amino acid composition, gap size, gap mass, sequence length, and hydrophobicity-based physicochemical descriptors, providing the richest set of features for this task to date.
- We develop a constraint-aware re-ranking approach that guarantees mass and biological compatibility of all generated candidates and greatly improves prediction accuracy for the known-mass variant.

This paper is further structured as follows. Section 2 discusses previous work in scaffold gap filling, de novo peptide sequencing, and protein language models. Section 3 defines the data collection process and methodology. Section 4 outlines the results analysis, and discussion. Section 5 describes limitations, and error analysis. Section 6 includes a conclusion.

II. LITERATURE REVIEW

Protein scaffold gap filling is a problem in de novo protein sequencing, where the tandem mass spectrometry processing results in an incomplete amino acid sequence with one or more gaps. Badal et al. [8] established the "known gap size" variant - where the number of missing amino acids is known - and the "known gap mass" variant - where only the mass of the missing amino acids is known. For the former, traditional machine learning algorithms (kNN, decision tree, random forest) were trained on encoded features of the flanking context; for the latter, a probabilistic scoring algorithm of candidate sequences was introduced that selects sequences based on compatibility with the gap mass and frequency of homologous hits. The journal-length article [6] extended these methods with more sophisticated feature engineering, and also introduced the preprocessing techniques which created the standard benchmarking datasets (MabCampath, CAH2, P5A) for the line of research.

In another study, Muzaffarov et al. [9] addressed the scaffold filling problem for multiple gaps, where multiple disjoint gaps are to be filled simultaneously - a much more difficult combinatorial version that inspired the evaluation with multiple gaps in the proposed system. Sturtz et al. [10] presented the first deep learning model for scaffold filling - a convolutional denoising autoencoder (CDA) - which pioneers neural reconstruction as an approach to the problem. Subsequently, Qingge et al. [11] tested a family of generative models including transformer encoder-decoders and GPT-2, and found that GPT-2 filled gaps with 100% accuracy on the MabCampath benchmark, proving that autoregressive language modeling is a very effective paradigm for local sequence completion. Branch C of the hybrid model benefits from this discovery. More broadly in de novo protein sequencing, Klapproth-Andrade et al. [12] developed Spectralis, a CNN-based approach that classifies

fragment ion series in mass spectra to attain greater than 40% sensitivity at 90% precision (almost doubling the previous state-of-the-art) by integrating amino acid mass constraints into model design. Likewise, Ebrahimi and Guo [13] proposed Transformer-DIA for data-independent acquisition (DIA) MS data, achieving amino acid-level precision up to 34.8% higher than previous methods [14]. Gueto-Tettay et al. [15] showed training deep learning models on multi-enzyme peptide digests results in 100% sequence coverage for 8 of 10 antibody chains, which is conceptually similar to the proposed ensemble approach, where models with different inductive biases cover each other's blind spots.

Pretrained protein language models (pLMs) are powerful sequence encoders. Lin et al.'s ESM-2 [13] models, trained on more than 250 million sequences through masked language modeling, yield embeddings that resemble evolutionary and structural relationships, but without structural information. Elnaggar et al.'s ProtTrans [16] also trained BERT- and T5-based pLMs, used extensively for transfer learning in sequence prediction. Both can be used for deep feature extraction in the proposed system's pre-processing stage. RAG-ESM [17] is notable for augmenting ESM-2 with retrieval-augmented generation (RAG), using a set of homologous sequences to condition masked amino acid prediction during inference - with higher accuracy than much larger models, and new state-of-the-art results for motif scaffolding. This justifies the proposed system's use of NCBI BLAST homologs as the main source of evidence for probabilistic scoring and reranking. Qin et al. [11] offer an overview of deep learning methods for protein structure, from early neural networks to AlphaFold, and the shift from engineered to learned features.

The reviewed papers show the evolution of the field: ML and probabilistic methods [6], [8] defined the problem and set the initial bar; CNN and AE-based methods [10] introduced deep learning to the problem; and generative approaches [11] are now reaching near-perfect performance on single-protein datasets. The success of mass spectrometry-based sequencing [12], [14], [15] illustrates the usefulness of mass-constraint-compliant architectures, and pLMs [17], [18] and retrieval-augmented systems [17] deliver deep sequence encodings and homology-assisted prediction, respectively. Yet, no previous work integrates the three approaches - ML, probabilistic mass-constrained reasoning, and deep sequence modeling - into a single hybrid system with a learned fusion layer, nor has robustness against noisy masses, incomplete homologs, and large gaps been explored [9]. These issues are addressed by the suggested hybrid system.

III. METHODOLOGY

A. Data Collection

The experimental dataset for this study was generated from the Bioinformatics Protein Dataset - Simulated, a publicly available Kaggle dataset [19]. The dataset consists of 20000 synthetic protein sequences, each calculated for several physicochemical properties, including molecular weight, isoelectric

point, molar hydrophobicity score, total charge (at pH=7), polar and nonpolar proportions, sequence length, and functional class labels. The dataset was created for protein classification and exploratory machine learning, comprising 16,000 training samples and 4,000 test samples.

Because the original dataset was not designed to fill gaps in protein scaffolds, it has been modified for this study. A summary of the generated protein scaffold Dataset is presented in Table II. A contiguous subsequence of 3–12 amino acids was removed from each sequence to simulate a missing scaffold region present for each valid protein. This deleted fragment was used to represent the ground-truth gap sequence and all recorded details included its position (gap-position), size (gap-size), (monoisotopic-mass), left and right nearby contexts. Gap difficulty distribution is provided in Table III. This process produced two scaffold formats, a known-gap-size scaffold and a known-gap-mass scaffold (mass token). The final modified files were used to train and evaluate the proposed hybrid scaffold gap-filling framework. Table I provides the description of the protein scaffold gap-filling dataset created in this study.

All sequences are composed exclusively of the 20 canonical amino acids:

$$\{A, C, D, E, F, G, H, I, K, L, M, N, P, Q, R, S, T, V, W, Y\} \quad (1)$$

The dataset is fully synthetic and simulated, designed to mimic realistic protein length distributions while enabling controlled ground-truth evaluation. No external biological repositories such as UniProt or NCBI were incorporated, thereby preserving the benchmark’s controlled nature. A Work follow diagram of the proposed Architecture is presented in Fig. 1.

B. Data Preprocessing

In this study, a systematic pre-processing pipeline was implemented to ensure data quality and suitability for gap-filling tasks. The sequence column was automatically identified and extracted. All records underwent rigorous alphabet validation against the canonical 20-amino-acid set, and sequences containing invalid symbols or zero-length entries were discarded. To ensure sufficient flanking context for scaffold generation, only sequences with a minimum length of 30 residues were retained.

1) *Scaffold and Gap Sample Generation*: A deterministic scaffold-generation process was developed to simulate realistic protein gap-filling scenarios. For reproducibility, fixed random seeds of 42 were applied to both Python’s `random` module and NumPy. For each eligible sequence, a gap length g was uniformly sampled from

$$g \in [3, 12] \quad (2)$$

A valid insertion position was then selected such that at least 11 residues of context remained on both flanking sides. Each sequence was partitioned into left context, true gap sequence, and right context. Two scaffold representations were generated:

TABLE I: Description of the Protein Scaffold Gap-Filling Dataset

Attribute	Description
Dataset source	Synthetic protein sequence dataset obtained from Kaggle
Dataset type	Simulated protein scaffold gap-filling dataset
Total samples	20,000 scaffold samples
Training samples	16,000 samples
Testing samples	4,000 samples
Split ratio	80% training and 20% testing
Sequence length range	50–300 amino acids
Average sequence length	174.22 amino acids
Amino acid alphabet	20 canonical amino acids: A, C, D, E, F, G, H, I, K, L, M, N, P, Q, R, S, T, V, W, Y
Gap size range	3–12 amino acids
Average gap size	7.50 amino acids
Gap mass range	213.1113–1769.8619 Da
Average gap mass	890.37 Da
Scaffold representations	Known-gap-size scaffold and known-gap-mass scaffold
Known-gap-size representation	Missing region replaced by dash symbols according to the gap length
Known-gap-mass representation	Missing region replaced by a mass token representing the monoisotopic mass of the removed sequence
Mass tolerance	± 0.5 Da during mass-constrained candidate retrieval
Class labels	Enzima, Estrutural, Transporte, Receptora, and Outras

TABLE II: Summary of the Generated Protein Scaffold Dataset

Dataset	Samples	Seq. Length	Gap Size
Training set	16,000	50–300	3–12
Testing set	4,000	50–300	3–12
Total	20,000	50–300	3–12

TABLE III: Gap Difficulty Distribution in the Generated Dataset

Difficulty	Definition	Train	Test
Small	≤ 4 residues	3,229	801
Medium	5–8 residues	6,272	1,601
Large	> 8 residues	6,499	1,598

- 1) Known-gap-size scaffold: The gap was replaced with exactly g dash characters.
- 2) Known-gap-mass scaffold: The gap was replaced with a mass token of the form $[MASS_m]$, where the monoisotopic peptide mass of the removed gap was computed as using standard residue masses rounded to four decimal places.

$$m = \sum_{i=1}^g \text{mass}(aa_i) \quad (3)$$

Each generated sample was enriched with metadata including protein identifier, class label, original sequence length, gap coordinates, gap size, gap mass, difficulty category (small: ≤ 4 , medium: 5–8, large: > 8), and both scaffold strings.

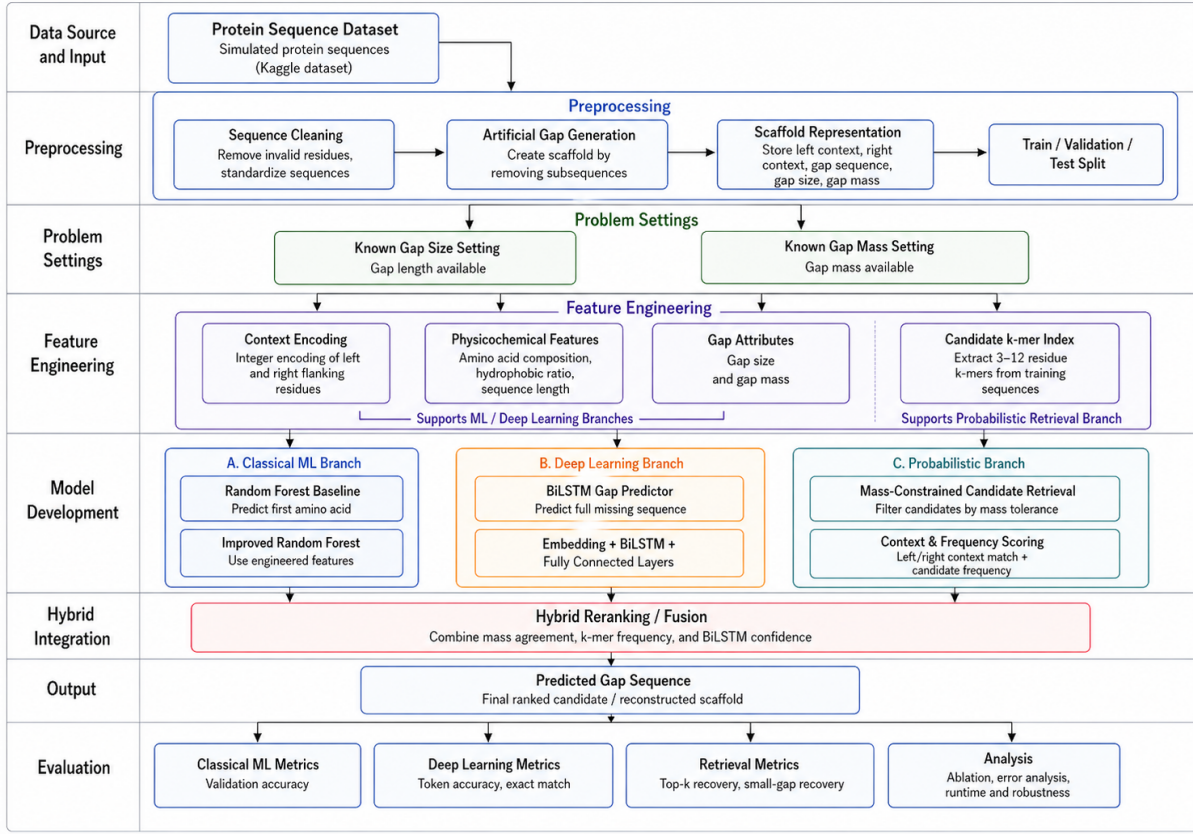


Fig. 1: Work follow diagram of the proposed Architecture

The processed datasets were saved as *known_gap_train.csv* and *known_gap_test.csv*.

C. Feature Engineering

For the classical machine learning branch, extensive feature engineering was performed. A custom feature extraction procedure generated structural descriptors, compositional statistics, and physicochemical properties.

Structural features included gap size, gap mass, and original sequence length. Compositional features consisted of raw counts of the 20 amino acids in the combined left-right context window of 22 residues. Additionally, hydrophobicity was defined as

$$HR = \frac{1}{|C|} \sum_{a \in C} \mathbb{I}(a \in \{A, I, L, M, F, W, Y, V\}) \quad (4)$$

where $\mathbb{I}(\cdot)$ denotes the indicator function.

Context sequences were integer encoded using

$$aa_to_idx = \{A : 1, C : 2, \dots, Y : 20, PAD : 0\} \quad (5)$$

and concatenated with engineered features to produce a final feature matrix of dimension $N \times 45$. In contrast, the deep learning and probabilistic branches operated directly on integer-encoded sequences without manual feature engineering.

D. Model Development

A hybrid modeling framework integrating three complementary paradigms was developed.

1) *Machine Learning Branch*: A Random Forest classifier was trained to predict the first amino acid of each gap. Two variants were constructed baseline model using encoded context only, and Enhanced model incorporating engineered features. This branch provided a lightweight benchmark for rapid inference.

2) *Deep Learning Branch*: For sequence-level prediction, a bidirectional long short-term memory architecture termed *GapPredictor* was implemented in PyTorch. The network consisted of an embedding layer (vocabulary size 21, embedding dimension 64), followed by a bidirectional LSTM with hidden dimension 128, and two fully connected layers with ReLU activation and dropout (0.2), producing logits of dimension

$$(B, 12, 21) \quad (6)$$

A specialized *SmallGapPredictor* model was additionally trained for gaps of length ≤ 4 .

Training minimized cross-entropy [20] loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{t=1}^T y_{i,t} \log(\hat{y}_{i,t}) \quad (7)$$

TABLE IV: Model Hyperparameters

Model	Hyperparameters
Random Forest (Baseline)	Number of Estimators: 100
	Maximum Depth: 20
	Random State: 42
Random Forest (Enhanced)	Number of Estimators: 200
	Maximum Depth: 25
	Random State: 42
Deep Learning (BiLSTM)	Embedding Dimension: 64
	Hidden Dimension: 128
	Dropout: 0.2
	Optimizer: Adam
	Learning Rate: 0.001
	Batch Size: 64

where T denotes padded gap length and PAD tokens were ignored using `ignore_index=0`.

3) Probabilistic Mass-Constrained Candidate Retrieval:

An exhaustive k -mer index spanning lengths 3–12 was constructed from training sequences, each indexed with its exact monoisotopic mass. During inference, candidates with matching gap length and mass were retrieved and ranked using a composite score based on Mass similarity, Left/right 3-mer context overlap and k -mer frequency in the training corpus, which provided biologically constrained candidate generation.

$$\pm 0.5 \text{ Da} \quad (8)$$

4) *Hybrid Reranker*: A hybrid reranking module rescored the top retrieved candidates using a weighted combination of mass agreement, frequency, and BiLSTM confidence. This integrated symbolic mass constraints with neural sequence probabilities.

$$\text{hybrid_score} = -2.0\Delta_{\text{mass}} + 0.3(\text{freq}) + 1.5(\text{DL_logprob}) \quad (9)$$

E. Training Procedure and Hyperparameter Selection

Hyperparameters were selected empirically within Kaggle notebook computational constraints. Table IV presents the hyperparameter settings.

The full-gap model was trained for 50 epochs, whereas the small-gap model was trained for 100 epochs. Stratified validation splitting used 20% validation data with random state 42. No formal early stopping was applied; convergence was monitored using loss and accuracy trajectories. For probabilistic and hybrid branches, mass tolerance was fixed. All stochastic operations were seeded at 42 for reproducibility.

$$\pm 0.5 \text{ Da} \quad (10)$$

$$k \in \{1, 3, 5\} \quad (11)$$

F. Performance Evaluation

Performance was assessed using task-specific metrics on held-out test scaffolds. For the machine learning branch, evaluation used validation accuracy and classification reports for first-residue prediction. For deep learning models, token-level accuracy was defined as below:

$$\text{token_acc} = \frac{\sum (\hat{y}_{i,t} = y_{i,t}) \cdot \mathbb{I}(y_{i,t} \neq 0)}{\sum \mathbb{I}(y_{i,t} \neq 0)} \quad (12)$$

Additional metrics included exact-match accuracy and mean residue accuracy. For probabilistic and hybrid branches, performance was measured using top- k recovery rates, representing the proportion of samples where the true gap appeared among the top- k ranked candidates, stratified by gap difficulty. Training and validation loss and accuracy curves were plotted to assess convergence and generalization. Summary statistics were exported to CSV for downstream analysis and visualization.

Overall, this integrated methodology enables systematic comparison of classical machine learning for rapid inference, deep learning for end-to-end sequence modeling, and probabilistic mass-constrained reasoning, providing a comprehensive framework for protein scaffold gap filling under controlled known-size and known-mass constraints.

IV. EXPERIMENTAL RESULTS

This section shows the experimental results of the proposed hybrid protein scaffold gap-filling framework. Evaluation was performed on the scaffold test set, which had been held out during scaffold generative model training, and generated from the simulated protein sequence dataset. The following experiments were conducted to benchmark four key components: the baseline Random Forest model, the enhanced Random Forest model with engineered features, the BiLSTM-based deep learning model, and the probabilistic mass-sandwiched candidate retrieval with hybrid rerank.

A. Classical Machine Learning Results

First experiment to evaluate the classical machine learning approach for first-residue prediction. Here, the model was trained to predict the first amino acid in a gap (shaded residue) based on either the left or right flanking context. The baseline Random Forest model (B-Model) included only integer-encoded context residues as input features. In contrast, the improved Random Forest model (I-Model) included additional features: gap size, gap mass, sequence length, and physicochemical properties based on amino acid composition and hydrophobicity. The performance Summary of the Proposed Scaffold Gap-Filling Framework is presented in Table V.

The validation accuracy for the baseline Random Forest was 0.0497, and it improved to 0.0628 in the final version! As there are 20 canonical amino acids, random predictions would be predicted to have an accuracy of around 0.05. Hence, the baseline model was close to random chance, while the feature-enhanced scores were slightly better. This shows that engineered gap features, such as gap mass, amino acid

TABLE V: Performance Summary of the Proposed Scaffold Gap-Filling Framework

Model / Branch	Task	Metric	Score
Random Forest Baseline	First amino acid prediction	Validation Accuracy	0.0497
Improved Random Forest	First amino acid prediction	Validation Accuracy	0.0628
BiLSTM Model	Full gap prediction	Token Accuracy	0.0501
Probabilistic Candidate Model	Mass-based candidate recovery	Small-gap Top-5 Recovery	0.0930
Hybrid Reranker	Deep learning + probabilistic reranking	Overall Top-5 Recovery	0.0137
Hybrid Reranker	Deep learning + probabilistic reranking	Small-gap Top-5 Recovery	0.0930

composition, and overall hydrophobicity, do help a little. However, the local context still lacks sufficient information for accurate gap reconstruction.

B. Deep Learning Results

The deep learning branch based on BiLSTM was further assessed for full-gap sequence forecasting. Unlike the Random Forest models, which could only predict the first amino acid of the gap, our BiLSTM model attempted to recover the full missing sequence, with a maximum length of 12 residues. An embedding layer, followed by a bidirectional LSTM layer, and then fully connected layers were used to make position-wise amino acid predictions.

The BiLSTM yielded a token-level validation accuracy of 0.0501. As with the previous tests, the token-level result was again close to the random amino acid baseline. The exact-match accuracy of the reconstructed sequences over the missing gap sequence was zero on the test set, indicating that no complete missing gap sequence had been perfectly reconstructed.

This finding indicates that there is too little learnable local sequence signal in the synthetic scaffold gaps for the flanking sequence context to recover all of the gaps via flap. Exact-match prediction is especially difficult for long gaps, since the model must accurately predict every residue in the missing subsequence. Standard training procedures cannot convert inference loss into prediction accuracy, as small fact-level mistakes can lead to infeasible predictions across complete sequences.

C. Probabilistic Candidate Retrieval Results

Firstly, the probabilistic branch, which scores candidates based on three features, was tested in a setting where the number of candidates to be retrieved is constrained, given their mass. For gap lengths in the range of 3–12 residues, a k-mer index was built from the training sequences. Inference filtered target gap mass (± 0.5 Da) and ranked candidate sequences using a combination of mass similarity, left and right context compatibility, and frequency in the training corpus.

For comparison, the probabilistic candidate retrieval model obtained a small-gap Top-5 recovery score of 0.0930. Thus,

for small gaps, the actual missing sequence lay in the top 5 retrieved candidates in about 9.30% of cases examined. This score, while still somewhat one-dimensional, is higher than those achieved by the ML and BiLSTM in direct predictions. This is because mass-constrained candidate filtering reduces the search space and, in some cases, recovers the true gap when the missing sequence is short.

On the other hand, for medium and large gaps, recovery performance was poor. Our suspicion is confirmed: the amino acid composition for gap lengths of 2 or greater increases exponentially. Also, because many amino acid sequences can have similar or identical masses, using only mass alone for filtering is not particularly unambiguous.

D. Hybrid Reranking Results

The hybrid reranker merged the outputs of the probabilistic retrieval branch and the BiLSTM model. The candidates first pass through the mass-constrained retrieval stage. Each candidate was subsequently scored with a weighted hybrid score based on mass difference, k-mer frequency, and deep learning log-probability.

The hybrid reranker obtains an overall Top-5 recovery score of 0.0137 and a small-gap Top-5 recovery score of 0.0930. The small-gap recovery remained similar to that of the probabilistic candidate branch, while the overall recovery was low due to poor performance on medium and large gaps.

These results indicate that hybrid reranking can retain useful small-gap candidates, but is ineffective when the deep learning model has weak predictive confidence. The BiLSTM branch yielded token-level accuracy close to random, so its score did not significantly boost candidate ranking. Thus, while the hybrid framework is structurally sound, it requires more robust biological features to improve its practical performance (e.g., homologous sequences, conservation scores, or embeddings from pretrained protein language models).

The data reveal that filling gaps in protein scaffolds is a daunting challenge within the current synthetic experimental regime. The empirical performance of classical machine learning and deep learning models was near-chance, indicating that local context alone is inadequate for accurate sequence reconstruction. This tendency confirmed the useful hypothesis that mass filtering also worked well for small gaps, though the probabilistic mass-constrained branch beat it. The overall performance, however, was found to be quite limited, particularly at the junction of medium and large gaps.

V. ERROR ANALYSIS AND LIMITATIONS

The experimental results show that the proposed scaffold gap-filling system has several limitations. First, the models mainly depend on local left- and right-sequence context, but this context was often insufficient to identify the exact missing amino acid sequence. As a result, both the Random Forest and BiLSTM models produced low accuracy, with performance close to random amino acid prediction.

Second, the dataset used in this study is synthetic and does not include real homologous sequences from databases such

as UniProt or NCBI. Therefore, the probabilistic model could not use evolutionary conservation or homologous sequence support, which are important for biologically reliable gap filling. This limited the system’s ability to distinguish the true gap sequence from other possible candidates.

Third, the known-mass setting suffers from mass ambiguity. Different amino acid sequences can have very similar or identical total masses, so mass filtering alone cannot always identify the correct sequence. This problem becomes more serious for medium and large gaps because the number of possible candidate sequences increases rapidly with gap length.

Another limitation is that the current mass-based retrieval method uses both gap mass and gap length during candidate selection. Therefore, it represents a constrained known-mass experiment rather than a fully uncertain-size known-mass setting. Future work should search across multiple possible gap lengths using only mass, context, and candidate scores.

The proposed approach demonstrates the feasibility of combining machine learning, deep learning, and probabilistic mass-constrained scoring. However, better performance will require real homologous-sequence retrieval, conservation features, pretrained protein-language-model embeddings, and stronger candidate reranking methods.

VI. CONCLUSION

This research applied a hybrid model for gap filling in protein scaffolds with controlled, known size and value preconditions. This system combined classical machine learning, BiLSTM-based deep learning, mass-constrained candidate retrieval, and hybrid reranking. However, local flanking context alone was deemed insufficient for accurate exact sequence recovery on the synthetic dataset, as experimental results show that classical and neural models performed close to random amino acid prediction. However, the non-zero top-k recovery was enabled by mass filtering despite small gaps; the mass-constrained branch performed favorably. The hybrid reranking framework lays the groundwork for future improvements and, specifically via homologous sequences, conservation, and pretrained protein language model (PTM) embeddings, offers tremendous opportunities to fine-tune our models further.

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